

Jeremy Malloch

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Skills

Python, C++, PyTorch, Distributed Training, Agentic Evals, RL Infrastructure, ML Modelling & Deployment

Work Experience

Member of Technical Staff – Backflip AI *San Francisco, CA* **Nov 2025 – Present**

- Co-built headless parametric computer aided design (CAD) package with commercial geometry kernel and constraint solver, built **multi-turn RL environments** on top of this
- Rearchitected distributed training for stability and **4x higher effective-token MFU**, upstreamed Qwen3.5 varlen attention [to torchtitan \(merged\)](#)
- Improved dataset quality with quality filtering, part deduping, a synthetic engineering drawing generation pipeline, and measuring Kaplan style data scaling coefficients to evaluate the quality of CAD augmentation techniques
- Co-built frontier VLM CAD harness, diagnosed agent reward hacking, driving **60% cost reduction & 63% lower reconstruction error** through harness engineering
- Set engineering standards on junior heavy team (linting, type checking, concise PR culture) through mentorship and reviewing 168 PRs across 8 engineers

Senior Machine Learning Engineer – Cruise *San Francisco, CA* **July 2021 – Aug 2025**

- Unblocked Cruise's return to driverless operation by leading Unified Tracker V2 model release that resolved critical ultra-nearfield risks (largest perception risk area)
 - Worked with upstream detection teams to root cause false positive detections, mine & label new datasets, and jointly train detection models & tracking models
 - Model shipped within 4 weeks of the Unified Tracker V1 release, ahead of schedule, unblocking driverless operation by end of 2024
- Contributed multiple modelling improvements to novel Unified Tracker V1 multi-modal, temporal tracking transformer model:
 - Increased model flop utilization by **32%**
 - Led deployment, contributed C++ changes for **3x speedup** in on car node, coordinated with downstream consumers to resolve latency blockers
 - Stabilized training through investigating internal model activation magnitudes, which unblocked lower / mixed precision training
 - Reduced velocity flips 20%, **increased pedestrian recall 95.0% to 98.3%**
- Accelerated inference of on car deployed models (perception, planning) by developing in-house PyTorch inference optimization library (quantization, pruning) with **torch.fx**
- Led two engineer team to build torch.fx based compiler frontend for bit-accurate emulation of DNN inference on custom inference hardware
- Created tooling to diagnose train to deploy numerical mismatches

Autonomous Systems Software Intern - Apple *Sunnyvale, CA* **Jan 2019 – Aug 2019**

- Contributed to machine learning compiler/inference framework in **C++ & Python**
- Researched & prototyped DNN quantization & sparsification pipeline, resulting in up to **40% faster** runtime of deployed models

Education

University of Waterloo **Waterloo, ON**

BMath in Computer Science and Combinatorics & Optimization

2021

Relevant Courses: Real Analysis, Group Theory, Optimization (Convex + Linear Programming)